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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Music Store Data Analysis

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About me

- I'm KATRAVATH SANTHOSH , I had recently completed my Bachelor's Degree in Engineering from Bharat Institution of Engineering Technology from CSE(cs)
- Github : https://github.com/shravanrathod32048-collab
- LinkedIn : http://linkedin.com/in/katrvath-santhosh-63758a394
- "I chose Data Science because I like analyzing information, identifying trends, and finding solutions based on evidence."

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Agenda

- Business Problem
- Business Statement
- Introduction
- Data Description
- Queries
- Insights
- Recommendations
- Challenges & Experience
- Conclusion

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Business Problem

- The business has difficulty identifying high-value customers and understanding customer spending patterns.
- The business lacks clear visibility into which countries and cities are performing better in terms of invoice activity and sales.
- The business has limited understanding of customer genre preferences across different countries.
- The business lacks insight into which Rock artists have the highest number of tracks in its music catalog.
- The business has difficulty understanding customer spending across different artists and identifying artist-level sales performance.

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Business Statement

- **The music store needs to analyze its large volume of *customer, invoice, track, artist, album, and genre data* to understand *customer purchasing behavior* and *sales performance*.**
- The analysis focuses on identifying **high-value customers**, **top-performing countries and cities**, **popular music genres**, and **top Rock artists**.
- Using **SQL-based data analysis**, the business can convert raw transactional data into **meaningful business insights** to support **better sales and marketing decisions**.

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Introduction

- **Music Store Data Analysis** is a SQL-based data analysis project focused on analyzing a music store's **customers, invoices, tracks, artists, albums, and genres**. The project uses a relational database to organize and analyze large amounts of **music and sales data**. Using **SQL queries, joins, aggregations, subqueries, and ranking techniques**, the project identifies **customer spending patterns, sales performance, popular genres, and top-performing artists**. The analysis helps convert raw music store data into **meaningful business insights**.

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Data Description

About Tables

```sql
--- 1. Genre
CREATE TABLE Genre (
    genre_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(120)
);
select * from genre;
```

```sql
--- 2. MediaType
CREATE TABLE MediaType (
    media_type_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(120)
);
select * from  MediaType;
```

```sql
--- 3. Employee
CREATE TABLE Employee (
    employee_id INT PRIMARY KEY AUTO_INCREMENT,
    last_name VARCHAR(120),
    first_name VARCHAR(120),
    title VARCHAR(120),
    reports_to INT,
    levels VARCHAR(255),
    birthdate DATE,
    hire_date DATE,
    address VARCHAR(255),
    city VARCHAR(100),
    state VARCHAR(100),
    country VARCHAR(100),
    postal_code VARCHAR(20),
    phone VARCHAR(50),
    fax VARCHAR(50),
    email VARCHAR(100)
);
```

```sql
--- 4. Artist
CREATE TABLE Artist (
    artist_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(120)
);
```

```sql
--- 5. Playlist
CREATE TABLE Playlist (
    playlist_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(255)
);
```

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```sql
--- 6. Album
CREATE TABLE Album (
    album_id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(160),
    artist_id INT,
    FOREIGN KEY (artist_id)
        REFERENCES Artist(artist_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
select * from  Album;
```

```sql
--- 7. Track
CREATE TABLE Track (
    track_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(200),
    album_id INT,
    media_type_id INT,
    genre_id INT,
    composer VARCHAR(220),
    milliseconds INT,
    bytes INT,
    unit_price DECIMAL(10,2),

    FOREIGN KEY (album_id)
        REFERENCES Album(album_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,

    FOREIGN KEY (media_type_id)
        REFERENCES MediaType(media_type_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,

    FOREIGN KEY (genre_id)
        REFERENCES Genre(genre_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
```

```sql
--- 8. Customer
CREATE TABLE Customer (
    customer_id INT PRIMARY KEY AUTO_INCREMENT,
    first_name VARCHAR(120),
    last_name VARCHAR(120),
    company VARCHAR(120),
    address VARCHAR(255),
    city VARCHAR(100),
    state VARCHAR(100),
    country VARCHAR(100),
    postal_code VARCHAR(20),
    phone VARCHAR(50),
    fax VARCHAR(50),
    email VARCHAR(100),
    support_rep_id INT,

    FOREIGN KEY (support_rep_id)
        REFERENCES Employee(employee_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
```

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```sql
--- 9. Invoice
CREATE TABLE Invoice (
    invoice_id INT PRIMARY KEY AUTO_INCREMENT,
    customer_id INT,
    invoice_date DATE,
    billing_address VARCHAR(255),
    billing_city VARCHAR(100),
    billing_state VARCHAR(100),
    billing_country VARCHAR(100),
    billing_postal_code VARCHAR(20),
    total DECIMAL(10,2),

    FOREIGN KEY (customer_id)
        REFERENCES Customer(customer_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
```

```sql
--- 10. InvoiceLine
CREATE TABLE InvoiceLine (
    invoice_line_id INT PRIMARY KEY AUTO_INCREMENT,
    invoice_id INT,
    track_id INT,
    unit_price DECIMAL(10,2),
    quantity INT,

    FOREIGN KEY (invoice_id)
        REFERENCES Invoice(invoice_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,

    FOREIGN KEY (track_id)
        REFERENCES Track(track_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
```

```sql
--- 11. PlaylistTrack
CREATE TABLE PlaylistTrack (
    playlist_id INT,
    track_id INT,

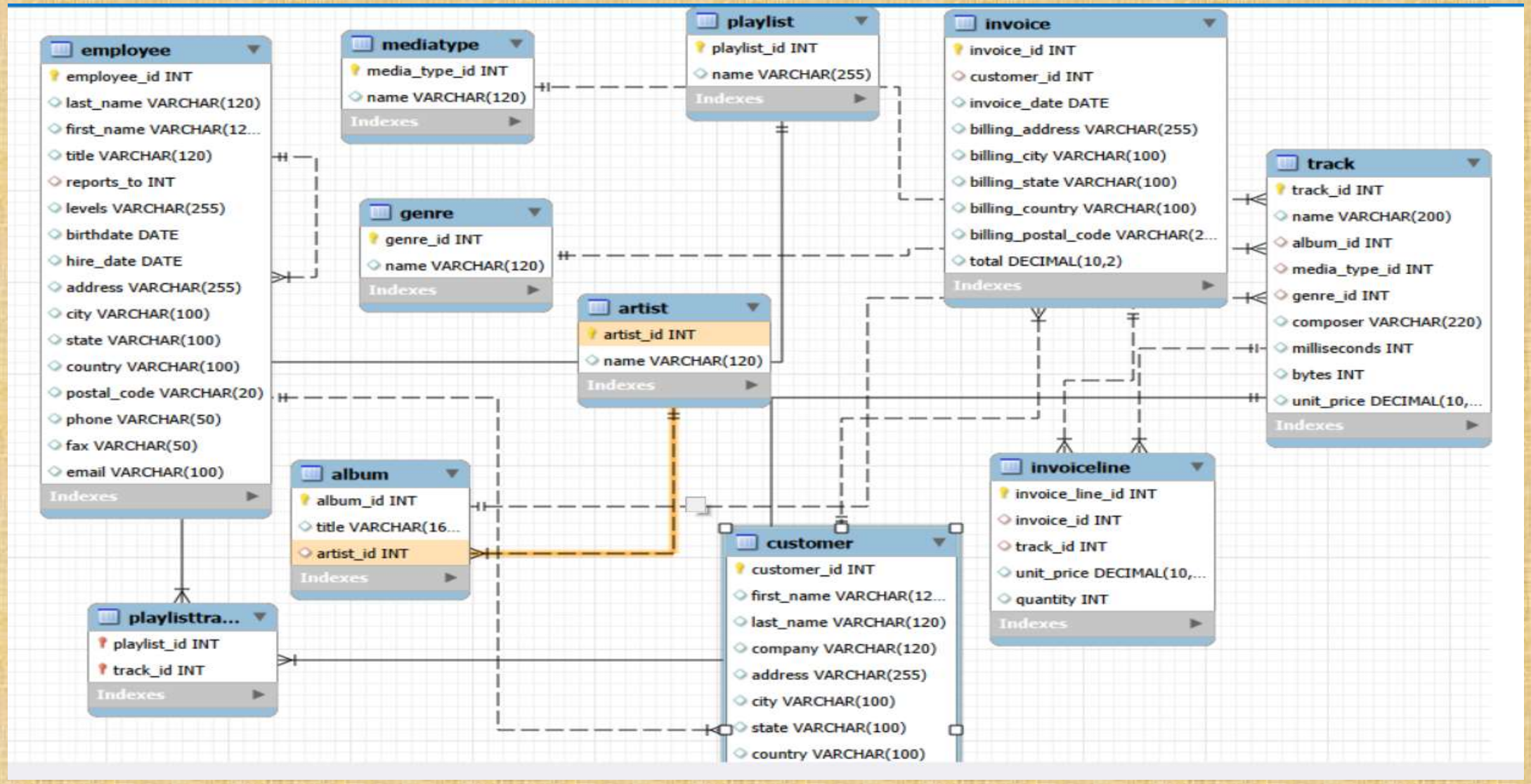
    PRIMARY KEY (playlist_id, track_id),

    FOREIGN KEY (playlist_id)
        REFERENCES Playlist(playlist_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE,

    FOREIGN KEY (track_id)
        REFERENCES Track(track_id)
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
```

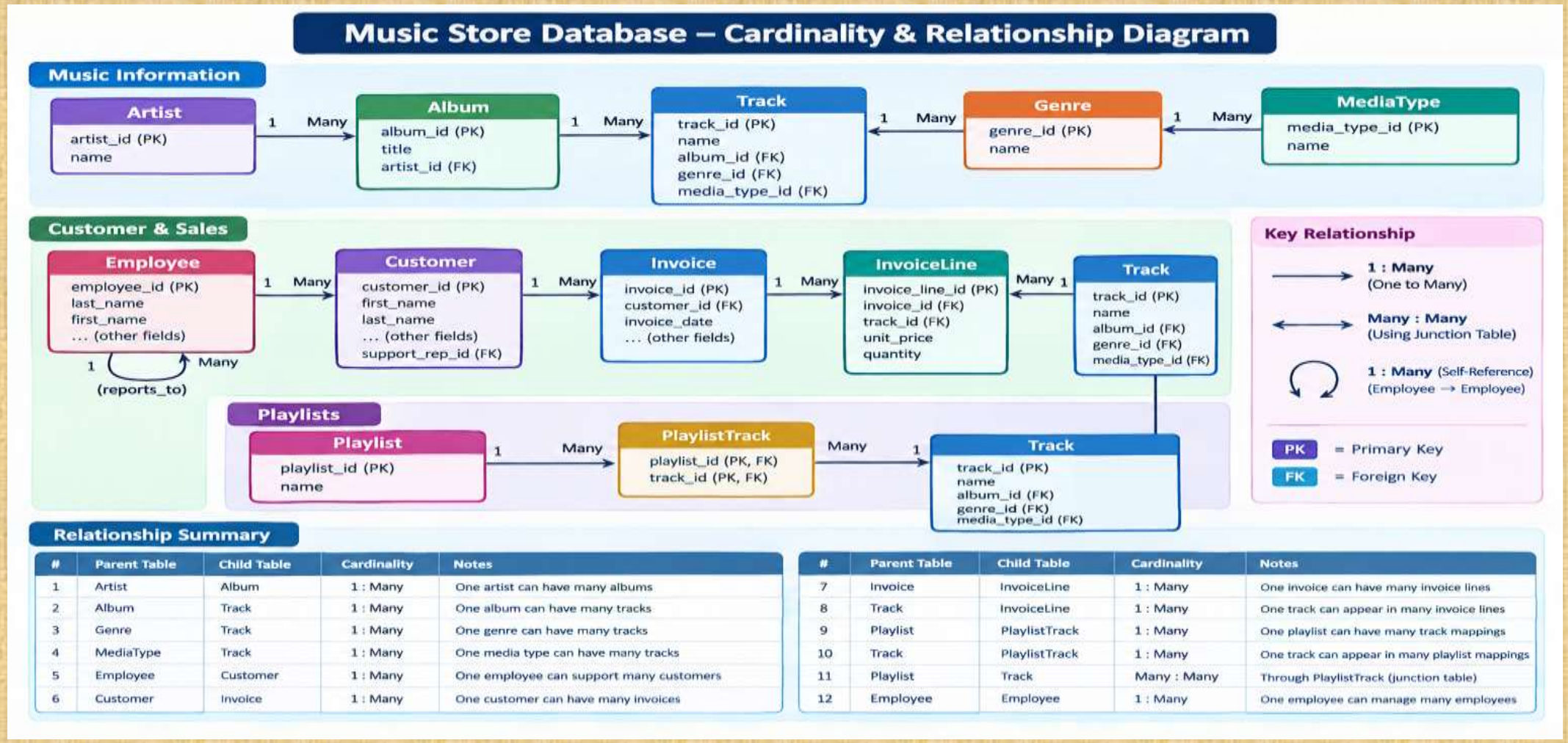
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ER Diagram & Schema Explanation



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CARDINALITIES



Relationship Summary

#	Parent Table	Child Table	Cardinality	Notes
1	Artist	Album	1 : Many	One artist can have many albums
2	Album	Track	1 : Many	One album can have many tracks
3	Genre	Track	1 : Many	One genre can have many tracks
4	MediaType	Track	1 : Many	One media type can have many tracks
5	Employee	Customer	1 : Many	One employee can support many customers
6	Customer	Invoice	1 : Many	One customer can have many invoices

#	Parent Table	Child Table	Cardinality	Notes
7	Invoice	InvoiceLine	1 : Many	One invoice can have many invoice lines
8	Track	InvoiceLine	1 : Many	One track can appear in many invoice lines
9	Playlist	PlaylistTrack	1 : Many	One playlist can have many track mappings
10	Track	PlaylistTrack	1 : Many	One track can appear in many playlist mappings
11	Playlist	Track	Many : Many	Through PlaylistTrack (junction table)
12	Employee	Employee	1 : Many	One employee can manage many employees

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Queries

1. Who is the senior most employee based on job title?

```sql
CREATE VIEW employee_levels AS
SELECT
    employee_id,
    first_name,
    last_name,
    title,
    levels
FROM Employee;
SELECT
    employee_id,
    first_name,
    last_name,
    title
FROM employee_levels
ORDER BY levels DESC
LIMIT 1;
```

employee_id	first_name	last_name	title
9	Mohan	Madan	Senior General Manager
NULL	NULL	NULL	NULL

The analysis identifies the employee with the **highest organizational level**,
helping the business understand its **management hierarchy and senior leadership structure**

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2. Which countries have the most Invoices?

```
DELIMITER //

CREATE PROCEDURE country_invoice_count()
BEGIN
    SELECT
        billing_country,
        COUNT(invoice_id) AS total_invoices
    FROM Invoice
    GROUP BY billing_country
    ORDER BY total_invoices DESC;
END //

DELIMITER ;
CALL country_invoice_count();
```

billing_country	total_invoices
USA	131
Canada	76
Canada	61
France	50
Germany	41
Czech Republic	30
Portugal	29
United Kingdom	28
India	21
Ireland	13

The analysis shows which countries generate the **highest invoice activity**, helping identify **strong customer markets** and locations with greater transaction volume.

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3. What are the top 3 values of total invoice?

```sql
WITH top_invoices AS
(
    SELECT
        invoice_id,
        total
    FROM Invoice
    ORDER BY total DESC
    LIMIT 3
)
SELECT
    invoice_id,
    total
FROM top_invoices;
```

invoice_id	total
183	23.76
92	19.80
31	19.80
NULL	NULL

Examining the highest invoice values helps the business understand its **largest individual transactions** and identify the upper range of customer purchases.

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4. Which city has the best customers? - We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals

```sql
SELECT
    billing_city,
    SUM(total) AS total_sales
FROM Invoice
GROUP BY billing_city
HAVING SUM(total) =
(
    SELECT MAX(total_sales)
    FROM
    (
        SELECT
            billing_city,
            SUM(total) AS total_sales
        FROM Invoice
        GROUP BY billing_city
    ) AS city_sales
);
```

	billing_city	total_sales
▶	Prague	273.24

Comparing invoice totals across cities reveals areas with **higher sales contribution**, helping the business understand **geographical sales performance**.

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5. Who is the best customer? - The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money

```sql
-- Q5. Find the customer who spent the most money
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    SUM(i.total) AS total_spent
FROM Customer c
JOIN Invoice i
    ON c.customer_id = i.customer_id
GROUP BY
    c.customer_id,
    c.first_name,
    c.last_name
ORDER BY total_spent DESC
LIMIT 1;
```

customer_id	first_name	last_name	total_spent
5	FrantiÅiek	WichterlovÃ¡	144.54

Identifying the highest-spending customer highlights **valuable customers**, which can help the business understand where **customer revenue is concentrated**.

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6. Write a query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A

```sql
-- Q6. Find customers who purchased Rock music
SELECT DISTINCT
    c.email,
    c.first_name,
    c.last_name,
    g.name AS genre
FROM Customer c
JOIN Invoice i
    ON c.customer_id = i.customer_id
JOIN InvoiceLine il
    ON i.invoice_id = il.invoice_id
JOIN Track t
    ON il.track_id = t.track_id
JOIN Genre g
    ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
ORDER BY c.email ASC;
```

email	first_name	last_name	genre
aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock
alero@uol.com.br	andre	Rocha	Rock
astrid.gruber@apple.at	Astrid	Gruber	Rock
bjorn.hansen@yahoo.no	BjÃ¸rn	Hansen	Rock
camille.bernard@yahoo.fr	Camille	Bernard	Rock
daan_peeters@apple.be	Daan	Peeters	Rock
diego.gutierrez@yahoo.ar	Diego	GutiÃ©rrez	Rock
dmiller@comcast.com	Dan	Miller	Rock
dominiquelefebvre@gmail.com	Dominique	Lefebvre	Rock
edfrancis@yachoo.ca	Edward	Francis	Rock

Analyzing customers who purchased Rock tracks reveals the level of **interest in the Rock genre**, providing insight into **customer music preferences**.

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7. Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands

```sql
-- Q7. Find the top 10 artists who have written the most Rock tracks
SELECT
    ar.name AS artist_name,
    COUNT(t.track_id) AS total_track_count
FROM Artist ar
JOIN Album al
    ON ar.artist_id = al.artist_id
JOIN Track t
    ON al.album_id = t.album_id
JOIN Genre g
    ON t.genre_id = g.genre_id
WHERE g.name = 'Rock'
GROUP BY
    ar.artist_id,
    ar.name
ORDER BY total_track_count DESC
LIMIT 10;
```

artist_name	total_track_count
Led Zeppelin	114
U2	112
Deep Purple	92
Iron Maiden	81
Pearl Jam	54
Van Halen	52
Queen	45
The Rolling Stones	41
Creedence Clearwater Revival	40
Kiss	35

Comparing Rock artists by track count identifies artists with a **large music catalog**, helping the business understand the **distribution of Rock content**.

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8. Return all the track names that have a song length longer than the average song length.- Return the Name and Milliseconds for each track. Order by the song length, with the longest songs listed first

```sql
-- Q8. Find tracks longer than the average song length
SELECT
    name,
    milliseconds
FROM Track
WHERE milliseconds >
(
    SELECT AVG(milliseconds)
    FROM Track
)
ORDER BY milliseconds DESC;
```

name	milliseconds
Occupation / Precipice	5286953
Through a Looking Glass	5088838
Greetings from Earth, Pt. 1	2960293
The Man With Nine Lives	2956998
Battlestar Galactica, Pt. 2	Battlestar Galactica
Battlestar Galactica, Pt. 1	2952702
Murder On the Rising Star	2935894
Battlestar Galactica, Pt. 3	2927802
Take the Celestra	2927677
Fire In Space	2926593
The Long Patrol	2925008
The Magnificent Warriors	2924716

Comparing track durations with the **average track length** helps the business understand the **characteristics of its music catalog** and identify unusually long tracks.

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9. Find how much amount is spent by each customer on artists? Write a query to return customer name, artist name and total spent

```sql
SELECT
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    ar.name AS artist_name,
    SUM(il.unit_price * il.quantity) AS total_spent
FROM Customer c
JOIN Invoice i
    ON c.customer_id = i.customer_id
JOIN InvoiceLine il
    ON i.invoice_id = il.invoice_id
JOIN Track t
    ON il.track_id = t.track_id
JOIN Album al
    ON t.album_id = al.album_id
JOIN Artist ar
    ON al.artist_id = ar.artist_id
GROUP BY
    c.customer_id,
    c.first_name,
c.last_name,
    ar.artist_id,
    ar.name
ORDER BY total_spent DESC;
```

customer_name	artist_name	total_spent
Hugh O'Reilly	Queen	27.72
Wyatt Girard	Frank Sinatra	23.76
FranÃ§ois Tremblay	The Who	19.80
Robert Brown	Creedence Clearwater Revival	19.80
FrantiÅ¡ek WichterlovÃ¡	Kiss	19.80
Helena HolÃ½	Red Hot Chili Peppers	19.80
Aaron Mitchell	James Brown	19.80
Heather Leacock	House Of Pain	18.81
Niklas SchrÃ¶der	Queen	18.81
Hugh O'Reilly	Nirvana	18.81
LuÃs GonÃ§alves	The Cult	17.82
FranÃ§ois Tremblay	Queen	17.82
Richard Cunningham	Marvin Gaye	17.82
FrantiÅ¡ek WichterlovÃ¡	Eric Clapton	17.82

Analyzing spending between customers and artists reveals **customer–artist purchasing patterns**, helping identify which artist content contributes to higher customer spending.

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10. We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared, return all Genres

```sql
SELECT
    country,
    customer_name,
    total_spent
FROM
(
    SELECT
        c.country,
        CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
        SUM(il.unit_price * il.quantity) AS total_spent,
        RANK() OVER (
            PARTITION BY c.country
            ORDER BY SUM(il.unit_price * il.quantity) DESC
        ) AS rnk
    FROM Customer c
    JOIN Invoice i
        ON c.customer_id = i.customer_id
    JOIN InvoiceLine il
        ON i.invoice_id = il.invoice_id
    GROUP BY
        i.billing_country,
        g.name
) AS x
WHERE rnk = 1
ORDER BY country;
```

country	genre	total_purchases
Argentina	Alternative & Punk	17
Australia	Rock	34
Austria	Rock	40
Belgium	Rock	26
Brazil	Rock	205
Canada	Rock	333
Chile	Rock	61
Czech Republic	Rock	143
Denmark	Rock	24
Finland	Rock	46
France	Rock	211
Germany	Rock	194
Hungary	Rock	44

Comparing genres across countries reveals **regional music preferences**, helping the business understand how **genre demand varies across markets**.

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11. Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount

```sql
SELECT
    country,
    customer_name,
    total_spent
FROM ranked_customers
WHERE customer_rank = 1
ORDER BY country;

-- Another Method
SELECT
    country,
    customer_name,
    total_spent
FROM
(
    SELECT
        c.country,
        CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
        SUM(il.unit_price * il.quantity) AS total_spent,
        RANK() OVER (
            PARTITION BY c.country
            ORDER BY SUM(il.unit_price * il.quantity) DESC
        ) AS rnk
    FROM Customer c
    JOIN Invoice i
        ON c.customer_id = i.customer_id
    JOIN InvoiceLine il
        ON i.invoice_id = il.invoice_id
    GROUP BY
        c.country,
        c.customer_id,
        c.first_name,
        c.last_name
) AS x
WHERE rnk = 1
ORDER BY country;
```

country	customer_name	total_spent
France	Wyatt Girard	99.99
Germany	Fynn Zimmermann	94.05
Hungary	Ladislav KovÃics	78.21
India	Manoj Pareek	111.87
Ireland	Hugh O'Reilly	114.84
Italy	Lucas Mancini	50.49
Netherlands	Johannes Van der Berg	65.34
Norway	BjÃ¸rn Hansen	72.27
Poland	StanisÅ‚aw WÃ³jcik	76.23
Portugal	JoÃ£o Fernandes	102.96
Spain	Enrique MuÃ±oz	98.01
Sweden	Joakim Johansson	75.24
United Kingdom	Phil Hughes	98.01
USA	Jack Smith	98.01

Identifying the highest spender within each country provides a **country-level view of customer value**, helping understand where individual customer spending is concentrated.

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Recommendations

- Focus marketing efforts on **high-value customers** and create **personalized offers** based on their purchasing patterns.
- Strengthen sales strategies in **high-performing countries and cities** to increase overall revenue.
- Promote **popular genres**, especially **Rock**, based on customer purchasing preferences across countries.
- Use **top-performing artists** to create targeted promotions, playlists, and artist-focused campaigns.
- Analyze **customer–artist spending patterns** to develop **personalized recommendations** and promotional offers.
- Monitor **invoice and sales trends** regularly using SQL to identify changes in **customer behavior and revenue performance**.
- Maintain accurate and consistent **customer, invoice, track, artist, and genre data** to support reliable business analysis.

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Challenges & Experience

- Faced challenges importing **large CSV datasets into MySQL** and gained practical experience in **database management and data importing**.
- Faced difficulties establishing **primary keys, foreign keys, and table relationships** and gained a clear understanding of **relational database design**.
- Faced challenges writing complex SQL queries using **JOINs, aggregate functions, subqueries, CTEs, and window functions**, improving **SQL problem-solving skills**.
- Encountered **SQL errors** during query development and gained experience in **debugging, troubleshooting, and validating query results**.
- Faced challenges analyzing **customer spending, sales performance, and country-wise genre preferences** and gained practical experience in **business data analysis**.
- Learned to transform **raw music store data into meaningful business insights and recommendations**, improving **analytical thinking and presentation skills**.

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Conclusion

- The **Music Store Data Analysis** project successfully demonstrates how **SQL** can be used to analyze a large relational dataset containing **customers, invoices, tracks, artists, albums, and genres**. The analysis identified important patterns in **customer spending, sales performance, genre preferences, and artist performance**. These insights can help the business understand its **customers and sales trends** and support more informed **marketing and sales decisions**.

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Any
Question

THANK YOU



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